Manendra Singh

Ratangarh, Churu, Rajasthan, 331022
Bachelor of Technology (B.Tech)
Computer Science and Engineering
Vellore Institute of Technology, Vellore

+91-7378221449
manendrasingh202005@gmail.com
GitHub Profile
LinkedIn Profile
Portfolio Website

Professional Summary

Computer Science undergraduate with strong expertise in Data Structures Algorithms, backend development, and scalable system design. Experienced in building secure and real-time applications using Node.js, MySQL, and REST APIs. Passionate about solving complex problems and developing high-performance software systems.

Education

•Vellore Institute of Technology, Vellore, Tamil Nadu *2023 - 2027*
Bachelor of Technology in Computer Science and Engineering CGPA: 8.55

•Swami Keshwanand Shikshan Sansthan, Sikar, Rajasthan *2021 - 2022*
Higher Secondary School (12th Standard) Percentage: 91.4

•Swami Keshwanand Shikshan Sansthan, Sikar, Rajasthan *2019 - 2020*
High School (10th Standard) Percentage: 88.33

Personal Projects

•Diabetic Retinopathy Detection(Live Demo) *GitHub*

An AI-driven medical diagnostic tool that classifies retinal images into five stages of diabetic retinopathy.

- Tech Stack: Python, TensorFlow, Keras, CNN, Streamlit, OpenCV, NumPy and PIL.
- Built a CNN model using TensorFlow/Keras to classify retinal images into five stages of diabetic retinopathy.
- Preprocessed medical images using OpenCV (resizing, normalization, augmentation) to improve model performance.
- Deployed the model via Streamlit to enable real-time medical image inference.

•SCSMS – Smart City Service Management System (Live Demo) *GitHub*

A scalable and real-time smart city platform for streamlined service reporting and citizen-admin communication.

- Tech Stack: Node.js, Express, MySQL (Aiven), Socket.IO, Leaflet.js, OpenStreetMap API, JavaScript.
- Built a scalable and secure backend system enabling real-time communication, reducing issue reporting latency by 40% using Socket.IO.
- Designed a secure relational database with role-based access control (RBAC) to ensure data integrity and efficient request resolution.
- Developed RESTful APIs following MVC architecture, ensuring maintainability, modularity, and adherence to software engineering best practices.

•Stack-IT (Live Demo) *GitHub*

A scalable community-driven discussion forum for developers to share technical knowledge through questions and answers.

- Tech Stack: MongoDB, Mongoose, Express.js, Node.js, JavaScript, HTML5, CSS3.
- Designed and deployed a scalable MERN-based platform with secure authentication and authorization mechanisms.
- Built RESTful APIs and optimized database queries to enhance performance, scalability, and efficient data retrieval.
- Developed a dynamic ranking algorithm based on user votes and engagement metrics to improve answer relevance and content discoverability.

Technical Skills and Interests

Programming Languages:Java, C++, Python, JavaScript
Machine Learning: TensorFlow, Keras, CNN, Scikit-learn, OpenCV, NumPy, Pandas
AI Concepts: Image Classification, Deep Learning, Model Evaluation, Data Preprocessing
Web Tech: Node.js, Express.js, React.js, HTML, CSS
Databases & Cloud: MongoDB, MySQL, Firebase (Cloud-based DB)
Core CS Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems
Software Engineering: REST APIs, MVC Architecture, Scalable Systems, Secure Coding, Debugging
Methodologies: Agile, CI/CD, Version Control (Git)
Developer Tools: Git, GitHub, Visual Studio Code

Achievements & Certifications

•LeetCode Solved 400+ algorithmic problems | Peak Rating: 1743 | Global Rank < 3,00,000 *Profile*

•GeeksforGeeks Resolved 350+ problems | Global Rank < 30,000 | Institute Rank < 500 *Profile*

•Complete Machine Learning & Data Science – Skill Up *GeeksforGeeks | Nation SkillUp*

•AI Agent Workshop using AgentForce *GeeksforGeeks & Salesforce – July 2025*

•Attended Google Cloud Agentic AI Day. *Hack2Skill – July 2025*

•Competed in Adobe India Hackathon *Adobe – July 2025*